Question ID: 358f18bc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$f(x) = x^2 - 48x + 2{,}304$$

What is the minimum value of the given function?

Correct Answer: 1728

Rationale

The correct answer is $1{,}728$. The given function can be rewritten in the form $f(x) = a(x-h)^2 + k$, where a is a positive constant and the minimum value, k, of the function occurs when the value of x is h. By completing the square, $f(x) = x^2 - 48x + 2{,}304$ can be written as $f(x) = x^2 - 48x + \left(\frac{48}{2}\right)^2 + 2{,}304 - \left(\frac{48}{2}\right)^2$, or $f(x) = (x-24)^2 + 1{,}728$. This equation is in the form $f(x) = a(x-h)^2 + k$, where $a = 1$, $h = 24$, and $k = 1{,}728$. Therefore, the minimum value of the given function is $1{,}728$.